

TIC 444465823

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

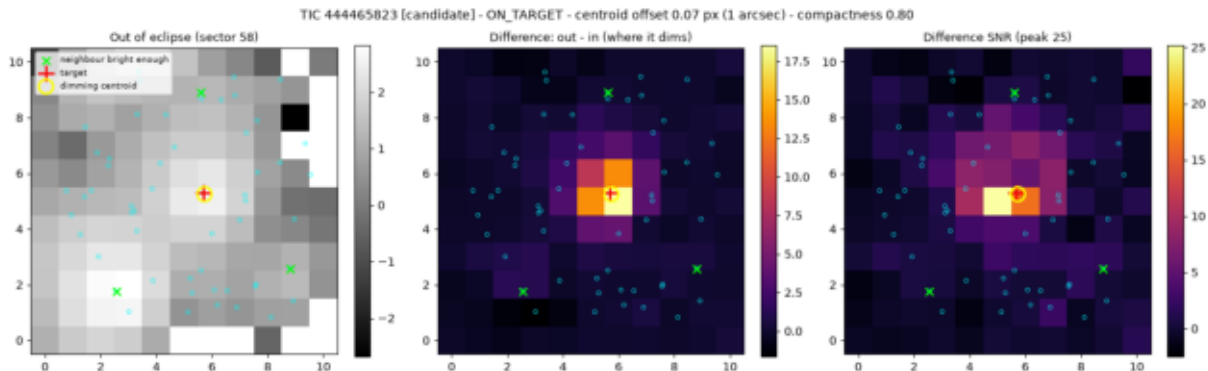
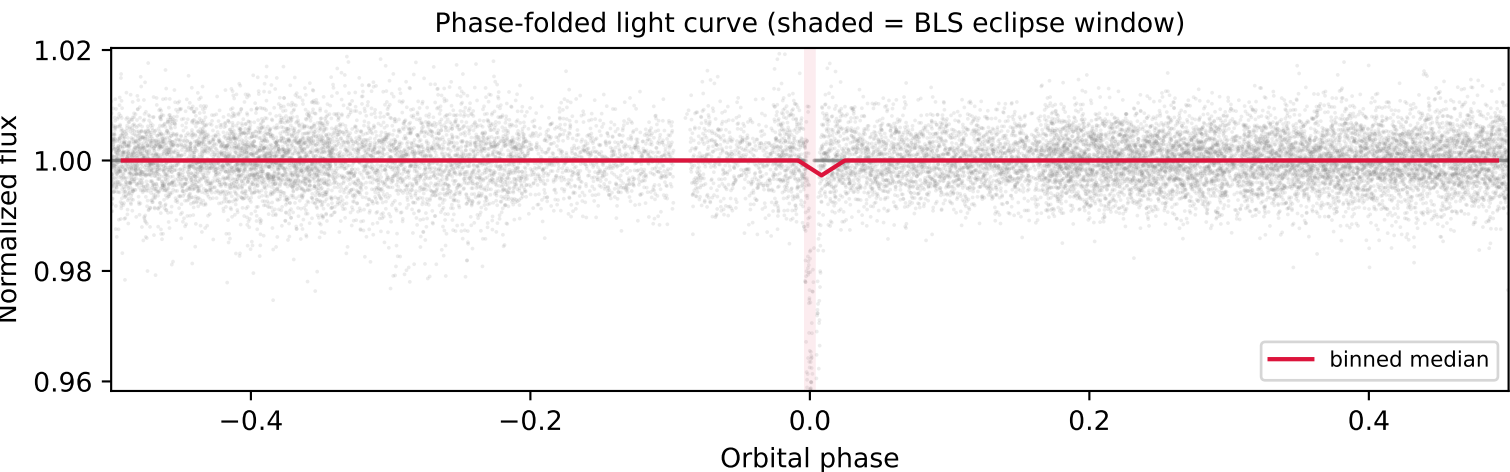
BLS EPHEMERIS
Period 18.513330 d (ALIASED, see below)
-> corrected period: 37.026660 d
Epoch (t0) 1768.5512 BTJD
Duration 0.1200 d
Depth 3.476 % (34,755 ppm)
SNR 146.6

CATALOG CROSS-MATCH (novelty)
VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

INDEPENDENT CHECKS (f3i gates)
Eclipses observed 8 (gate >= 3)
Robust depth ratio 1.01 (gate >= 0.50)
Robust depth sigma 95.7 (gate >= 7.0)
Odd/even z-score 14.96 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
Sector 58 light-curve dip in this sector: 61.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 25.18 (gate >= 5.0) Compactness 0.80 (gate >= 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 444465799
3.65 px away, dTmag +2.93 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 14.96 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 37.026660 d, in any write-up.